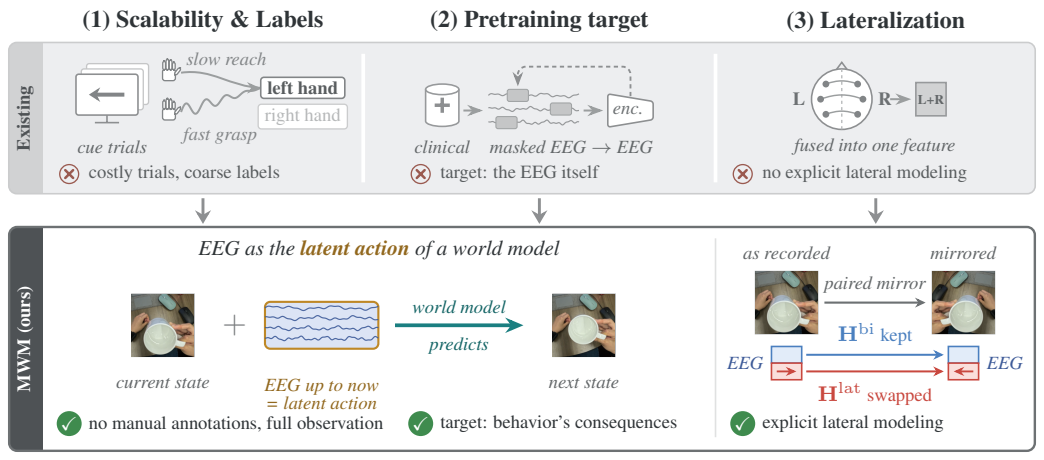

(1) Scalability & Labels
(2) Pretraining target
(3) Lateralization
Existing
slow reach
left hand
right hand
fast grasp
cue trials
costly trials, coarse labels
clinical
masked EEG → EEG
enc.
target: the EEG itself
L
R
L+R
fused into one feature
no explicit lateral modeling
MWM (ours)
EEG as the latent action of a world model
current state
EEG up to now
= latent action
world model
predicts
next state
no manual annotations, full observation
target: behavior's consequences
as recorded
mirrored
paired mirror
H^bi kept
EEG
EEG
H^lat swapped
explicit lateral modeling